

UBUNTU INNOVATIONS

(Pty) Ltd

IT Infrastructure Report

Week 1 — Foundations: Hardware & Networking Basics

Prepared by

Neliswa Mapisa

Aphelele Aplin

Lulama Vena

Lihle Jikela

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1. Business Requirements Analysis

1.1 Company Overview

Ubuntu Innovations (Pty) Ltd is a South African technology SME currently relocating to a new office. The company employs 25 full-time staff across five departments and requires a complete IT infrastructure to be designed, deployed, and documented from scratch to support day-to-day business operations.

As the Junior IT Support Specialist team, this report forms the first deliverable of a four-week build plan. It identifies who works at the company, what they do, what technology they need, and what risks need to be planned for.

Department	Headcount	Primary Function
Executives	3	Strategic leadership, board reporting, company governance
Finance	4	Accounting, payroll, budgeting, online banking
HR	4	Recruitment, employee records, compliance, training
Sales & Marketing	7	Client acquisition, campaigns, CRM, social media
Software Development	7	Product development, version control, cloud deployments
TOTAL	25	Full-spectrum SME operations

1.2 Departmental IT Requirements

1.2.1 Executives (3 staff)

The Executive team is responsible for strategic leadership, investor relations, and board-level reporting. Their IT requirements focus on mobility, secure access, and reliable communication tools.

Software needed:

- Microsoft 365 — Outlook, Word, Excel, PowerPoint, Teams
- Zoom or Microsoft Teams for remote meetings and board calls
- SharePoint or OneDrive for secure cloud document access
- Antivirus and BitLocker device encryption on all laptops

Hardware needed:

- 3 × premium business laptops (Dell Latitude or Lenovo ThinkPad)
- Docking stations and dual monitors for in-office use
- Smartphones enrolled in Mobile Device Management (MDM)

1.2.2 Finance Department (4 staff)

The Finance team handles all financial transactions, payroll, SARS tax submissions, and online banking. They work with highly sensitive data and require secure, regulated access to financial systems.

Software needed:

- Sage Accounting or Xero for bookkeeping and financial reporting
- Microsoft Excel (advanced) for budgets and reconciliations
- Secure browser access to online banking portals
- eFiling access (SARS) for VAT, PAYE, and income tax submissions

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- Adobe Acrobat for financial statements

Hardware needed:

- 4 × business desktops or laptops with dual monitors
- Dedicated network printer/scanner for physical financial documents
- UPS at every Finance workstation

1.2.3 Human Resources Department (4 staff)

The HR team manages recruitment, employee records, leave tracking, and compliance under the BCEA. They handle personally identifiable information and must comply with POPIA.

Software needed:

- HR management system: SimplePay, Sage HR, or BambooHR
- Microsoft 365 for document creation, email, and internal communication
- Secure digital filing system for employee contracts and ID documents
- Video conferencing for remote interviews (Teams or Zoom)

Hardware needed:

- 4 × business laptops or desktops
- Access to the shared office printer/scanner

1.2.4 Sales & Marketing Department (7 staff)

The Sales and Marketing team drives business growth through client acquisition, digital marketing campaigns, and CRM. They are the highest-traffic users of cloud tools and external web platforms.

Software needed:

- CRM platform: HubSpot, Salesforce, or Zoho CRM
- Email marketing: Mailchimp or HubSpot for campaigns and newsletters
- Social media management: Hootsuite or Buffer
- Design tools: Canva Pro or Adobe Express
- Microsoft 365 for proposals, presentations, and email

Hardware needed:

- 7 × laptops — mix of mid-range and higher-spec for designers
- Large monitors for design and presentation work
- Access to colour printer for marketing collateral

1.2.5 Software Development Department (7 staff)

The Software Development team builds, tests, and deploys Ubuntu Innovations' core products. They are the most technically demanding users, requiring high-spec hardware and secure access to cloud and version control environments.

Software needed:

- Version control: GitHub or GitLab for source code management
- Development IDEs: Visual Studio Code, IntelliJ IDEA, or PyCharm
- Containerisation: Docker Desktop and/or Kubernetes tools
- Cloud platforms: AWS, Azure, or Google Cloud Console
- Project management: Jira or GitHub Projects for sprint tracking

Hardware needed:

- 7 × high-spec workstations — min. 32 GB RAM, 1 TB NVMe SSD, multi-core CPU
- Dual or triple monitor setups per developer

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- Dedicated VLAN network segment separated from general staff

1.3 Shared IT Infrastructure Requirements

- High-speed fibre internet (minimum 100 Mbps symmetric) with 4G LTE failover router
- Managed network switch and enterprise Wi-Fi access points for all 25 users
- Windows Server or cloud-based Active Directory for centralised user management
- Network-attached storage (NAS) for shared drives and local backups
- Company-wide Microsoft 365 Business Standard licences for all staff
- Centralised antivirus management console (ESET Endpoint Security or Bitdefender GravityZone)
- UPS units in the server room and at Finance workstations

1.4 Risk Analysis

Risk / Threat	Mitigation Strategy	Severity
Power Outages (load-shedding)	UPS devices, surge protection, generator backup	HIGH
Internet Connectivity Loss	Dual ISP lines or LTE failover router	HIGH
Data Loss / Ransomware	Daily offsite backups, antivirus, staff training	CRITICAL
Unauthorised Access	Strong passwords, MFA, role-based permissions	HIGH
Hardware Failure	Warranty support, spare parts, cloud storage	MEDIUM
Staff IT Misuse	Acceptable use policy, content filtering	MEDIUM
Software Licensing Issues	Asset register, licensed software only	MEDIUM

1.5 Conclusion & Next Steps

Ubuntu Innovations (Pty) Ltd has clear and well-defined IT requirements across all five departments. The infrastructure must support 25 employees reliably, securely, and cost-effectively — from executive mobility through to developer workstations.

- Week 1 (current) — Procure hardware, set up network equipment, configure workstations
- Week 2 — Install operating systems, configure user accounts, set up shared drives
- Week 3 — Implement security policies, configure backups, deploy antivirus
- Week 4 — Final testing, staff onboarding, and handover documentation

2. Hardware Inventory

Total line items: 27 | Total units: 209 | Estimated total cost (excl. VAT): R 1,174,550 | Prepared: May 2026

2.1 Computing Equipment

Item	Description	Specifications	Dept	Qty	Unit (ZAR)	Total (ZAR)	Priority
COMP-01	Executive Laptop	Intel i7-1355U, 16 GB DDR5, 512 GB NVMe, Win 11 Pro	Exec	3	28,500	85,500	HIGH
COMP-02	Finance & HR Desktop	Intel i5-13400, 16 GB DDR4, 512 GB NVMe, Win 11 Pro	Fin, HR	8	14,500	116,000	HIGH
COMP-03	Sales & Mktg Laptop	Intel i5-1335U, 16 GB DDR4, 512 GB SSD, Win 11 Pro	Sales & Mktg	7	18,000	126,000	HIGH
COMP-04	Developer Workstation	i7-13700K, 32 GB DDR5, 1 TB NVMe, RTX 3060	Dev	7	38,000	266,000	HIGH

2.2 Monitors

Item	Description	Specifications	Dept	Qty	Unit (ZAR)	Total (ZAR)	Priority
MON-05	24" Full HD Monitor	1920x1080, IPS, 75 Hz, HDMI+DP	Fin, HR, Exec	16	4,200	67,200	HIGH
MON-06	27" QHD Monitor	2560x1440, IPS, 144 Hz, USB-C	Dev, Sales	10	7,500	75,000	MEDIUM

2.3 Networking Equipment

Item	Description	Specifications	Dept	Qty	Unit (ZAR)	Total (ZAR)	Priority
NET-07	Core Managed Switch	48-port Gigabit, PoE+, VLAN, QoS, rack-mount	Srv Room	1	18,000	18,000	CRITICAL
NET-08	Wi-Fi Access Point	Wi-Fi 6, dual-band, WPA3, PoE-powered	All Depts	4	5,500	22,000	HIGH
NET-09	Fibre Router/Firewall	1 Gbps WAN, VPN, stateful firewall, dual WAN	Srv Room	1	12,000	12,000	CRITICAL
NET-10	LTE Failover Router	4G LTE Cat-12, dual SIM, load balancing	ISP Backup	1	6,500	6,500	HIGH
NET-11	Cat6A Cabling (per pt)	30 data points, RJ45, patch panels	All Depts	30	850	25,500	HIGH

2.4 Servers & Storage

Item	Description	Specifications	Dept	Qty	Unit (ZAR)	Total (ZAR)	Priority
SRV-12	File & AD Server	Xeon E-2336, 32 GB ECC, RAID 1+5, Win Server 2022	Srv Room	1	55,000	55,000	CRITICAL
SRV-13	NAS Device	4-bay, 4x4 TB IronWolf, RAID 5, SMB/NFS	All	1	22,000	22,000	HIGH
SRV-14	Server Rack 12U	Lockable, vented, cable management	Srv Room	1	8,500	8,500	MEDIUM

2.5 Power Protection

Item	Description	Specifications	Dept	Qty	Unit (ZAR)	Total (ZAR)	Priority
PWR-15	Server Room UPS 3000VA	Pure sine wave, 8x IEC, SNMP, LCD display	Srv Room	1	18,000	18,000	CRITICAL
PWR-16	Workstation UPS 1200VA	Simulated sine, 4x outlet, USB shutdown	Finance, HR	6	2,800	16,800	HIGH
PWR-17	Surge Protector Strip	6-outlet, 2500 joule, indicator light	All Depts	15	380	5,700	MEDIUM

2.6 Peripherals, Printing & Security

Item	Description	Specifications	Dept	Qty	Unit (ZAR)	Total (ZAR)	Priority
PER-18	Wireless Keyboard & Mouse	Full-size, USB nano receiver, 12-month battery	All Depts	25	650	16,250	MEDIUM
PER-19	USB-C Docking Station	2x HDMI, 4x USB-A, Gigabit LAN, 100W PD	Exec, Sales	10	2,200	22,000	MEDIUM
PER-20	USB Wired Headset	Noise-cancelling mic, USB-A, Teams certified	All Depts	25	900	22,500	MEDIUM
PER-21	Full HD Webcam	1080p, auto-focus, privacy shutter	Exec, Sales	14	1,200	16,800	MEDIUM
PRT-22	Multifunction Laser MFP	A4/A3 colour laser, duplex, LAN+Wi-Fi, 500-sheet	Fin, HR, Sales	2	18,000	36,000	MEDIUM
PRT-23	Label Printer	Thermal, 300 dpi, USB+LAN	IT / HR	1	2,800	2,800	LOW
SEC-24	IP Camera System	4MP, 8-ch NVR, 2 TB HDD, PoE, night vision	Office	1	14,500	14,500	HIGH
SEC-25	Smart Door Access Control	RFID + PIN, 25x cards, audit log, 12V DC	Srv Room	1	6,500	6,500	HIGH
TEL-26	VoIP Desk Phone	SIP, 2-line, HD voice, PoE, Gigabit	Exec, Fin	10	2,500	25,000	MEDIUM
MOB-27	Company Android Phone	Android 14, 128 GB, 5G, enrolled in Intune MDM	Exec, Sales	7	9,500	66,500	MEDIUM

2.7 Hardware Budget Summary by Category

Category	Items	Units	Cost (ZAR)	% of Budget	Priority
Computing	4	25	593,500	50.5%	CRITICAL / HIGH
Monitors	2	26	142,200	12.1%	HIGH / MEDIUM
Networking	5	37	84,000	7.2%	CRITICAL / HIGH
Server & Storage	3	3	85,500	7.3%	CRITICAL / HIGH
Power Protection	3	22	40,500	3.4%	CRITICAL / HIGH
Peripherals	4	74	77,550	6.6%	MEDIUM
Printing	2	3	38,800	3.3%	MEDIUM / LOW
Security	2	2	21,000	1.8%	HIGH
Telephony	1	10	25,000	2.1%	MEDIUM
Mobile / MDM	1	7	66,500	5.7%	MEDIUM
TOTAL	27	209	1,174,550	100%	—

All prices are estimated retail/tender prices in South African Rand (ZAR) as at May 2026. Actual quotes may vary by 5–15%. VAT (15%) is excluded from all figures.

3. Software Inventory

Total licences: 232 | Estimated annual software cost (excl. VAT): R 638,440 | Last updated: May 2026

3.1 Human Resources

Software	Category	License Type	Licences	Cost (ZAR/yr)	Notes
Microsoft Office 365	Productivity Suite	Subscription	10	18,000	Word, Excel, Outlook, Teams
PaySpace Payroll	Payroll	Subscription	5	24,000	South African payroll & HR
BambooHR	HR Management	Subscription	3	15,000	Employee records & onboarding
Adobe Acrobat	Document Management	Subscription	5	9,000	PDF creation & editing

3.2 IT Department

Software	Category	License Type	Licences	Cost (ZAR/yr)	Notes
Microsoft 365 Admin	Administration	Included	2	0	Tenant management
ManageEngine ServiceDesk	Helpdesk / ITSM	Subscription	5	32,000	IT ticketing system
SolarWinds RMM	Network Monitoring	Subscription	3	28,000	Remote monitoring & management
Veeam Backup	Backup & Recovery	Perpetual	2	21,000	Server backup solution

3.3 Software Development

Software	Category	License Type	Licences	Cost (ZAR/yr)	Notes
Visual Studio Code	IDE / Code Editor	Free	15	0	Open source — no cost
JetBrains Suite	IDE	Subscription	10	45,000	IntelliJ, PyCharm, WebStorm
Git / GitHub Enterprise	Version Control	Subscription	15	38,000	Source code management
Docker Desktop	Containerisation	Subscription	15	22,000	Container runtime & tooling
Postman	API Testing	Subscription	8	9,600	API development & testing
Jira Software	Project Management	Subscription	15	18,000	Agile project tracking

3.4 Finance

Software	Category	License Type	Licences	Cost (ZAR/yr)	Notes
Microsoft Office 365	Productivity Suite	Subscription	8	14,400	Word, Excel, Outlook, Teams
Sage 300 ERP	Accounting / ERP	Perpetual	5	85,000	Financial management & reporting
Power BI Pro	Business Intelligence	Subscription	5	12,000	Financial dashboards & reports
Adobe Acrobat	Document Management	Subscription	4	7,200	PDF creation & editing

3.5 Sales & Marketing

Software	Category	License Type	Licences	Cost (ZAR/yr)	Notes
Microsoft Office 365	Productivity Suite	Subscription	12	21,600	Word, Excel, Outlook, Teams
Salesforce CRM	CRM	Subscription	12	72,000	Customer relationship management
HubSpot Marketing Hub	Marketing Automation	Subscription	4	36,000	Email campaigns & analytics
Canva for Teams	Design	Subscription	8	9,600	Marketing collateral & graphics
Zoom	Video Conferencing	Subscription	12	8,640	Client calls & webinars

3.6 Operations

Software	Category	License Type	Licences	Cost (ZAR/yr)	Notes
Microsoft Office 365	Productivity Suite	Subscription	8	14,400	Word, Excel, Outlook, Teams
Asana	Project Management	Subscription	8	14,400	Task & project tracking
Slack	Communication	Subscription	10	9,600	Internal team messaging
Microsoft Visio	Diagramming	Subscription	3	9,000	Process flow diagrams

3.7 Executive / Management

Software	Category	License Type	Licences	Cost (ZAR/yr)	Notes
Microsoft Office 365	Productivity Suite	Subscription	5	9,000	Word, Excel, Outlook, Teams
Power BI Pro	Business Intelligence	Subscription	5	12,000	Executive dashboards
DocuSign	E-Signature	Subscription	5	18,000	Digital contract signing

Software	Category	License Type	Licences	Cost (ZAR/yr)	Notes
LastPass Business	Password Management	Subscription	5	6,000	Secure credential management

3.8 Annual Software Cost Summary

Total estimated annual software spend: R 638,440 (232 licences across 7 departments). Excludes VAT, hardware, infrastructure, and installation. Open-source tools — VS Code and Git — are listed at R 0.

4. Network Topology Diagram

The diagram below shows the logical network topology for Ubuntu Innovations' new office. Internet traffic enters through the firewall appliance, passes through the core managed switch, and is distributed to four categories of endpoint devices: desktop computers, wireless access points, shared printers, and NAS storage.

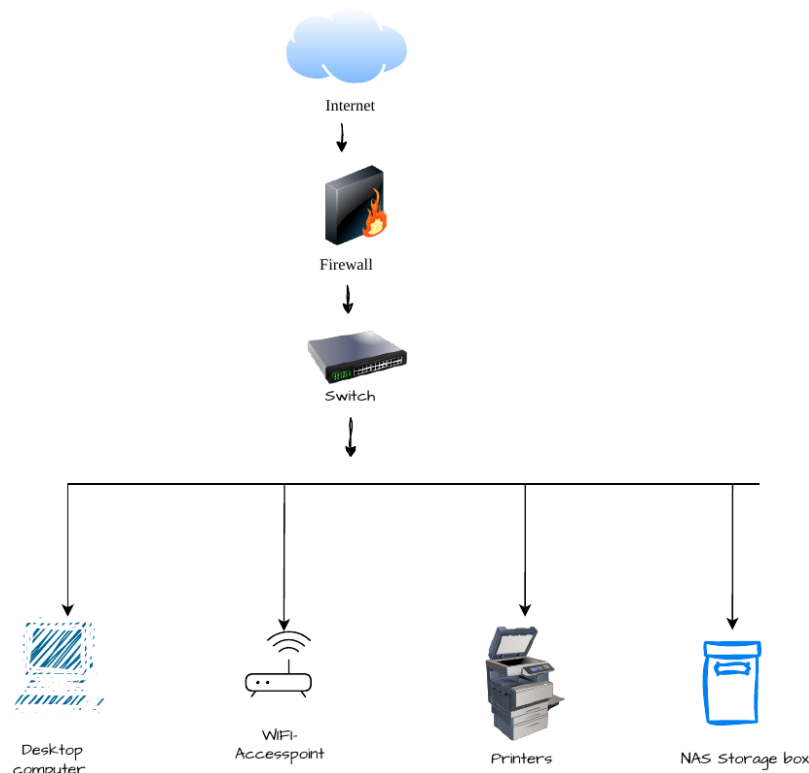


Figure 1 — Ubuntu Innovations Office Network Topology

4.1 Network Component Summary

Component	Description
Internet	External WAN via fibre ISP with 4G LTE failover
Firewall	FortiGate 60F or Ubiquiti Dream Machine — stateful inspection, VPN, dual WAN
Core Switch	Cisco SG350-48P — 48-port PoE+ managed switch with VLAN support
Desktop Computers	25 workstations and laptops across all five departments
Wi-Fi Access Points	4 x Ubiquiti UniFi U6-Pro (Wi-Fi 6) for full office wireless coverage
Printers	2 x Multifunction colour laser printers shared across Finance, HR, and Sales
NAS Storage	Synology DS923+ — 4-bay RAID 5 for shared drives and local backups

5. Troubleshooting Scenarios

This section documents step-by-step resolution procedures for the three most frequently reported IT issues across all departments. Each scenario includes a problem summary, affected systems, estimated resolution time, and a clear action plan. All IT support staff are expected to follow these procedures before escalating to Tier 2.

#	Scenario	Issue Type	Severity	Est. Resolution
1	Printer Offline	Connectivity / Hardware	Medium	15–30 min
2	Password Reset / Account Locked	Security / Access	High	5–10 min
3	Slow Computer Performance	Performance / Software	Low	30–60 min

5.1 Scenario 1 — Printer Offline

Severity: Medium | Est. Resolution: 15–30 minutes | Affects: All departments | Likely Causes: Power/cable issues, network, outdated drivers, queue corruption

#	Action	Details / Instructions
1	Check Power & Cables	Confirm the printer is switched on and the power cable is secure. Check the network cable or confirm Wi-Fi signal. Look for error lights on the control panel.
2	Restart the Printer	Power off completely, wait 30 seconds, then power back on. Wait for full initialisation before proceeding.
3	Verify Network Connection	Print a configuration page (hold Info button 3 sec) and confirm the printer shows a valid IP address on the same subnet as the user's PC.
4	Clear the Print Queue	Control Panel > Devices & Printers > right-click printer > See what's printing > Cancel all. If stuck: stop Print Spooler in services.msc, delete files in C:\Windows\System32\spool\PRINTERS, restart the service.
5	Set Printer Online	Right-click printer > See what's printing > Printer menu. Uncheck 'Use Printer Offline' if ticked.
6	Reinstall / Update Driver	Right-click printer > Remove device. Download the latest driver from the manufacturer's website and reinstall, or use Windows Update > Optional updates.
7	Test Print	Right-click printer > Printer properties > Print Test Page. Confirm the page prints correctly.
8	Escalate if Unresolved	Log a ticket in ManageEngine ServiceDesk with the printer IP, model, and a screenshot of any error. Assign to Tier 2 — Network/Hardware team.

5.2 Scenario 2 — Password Reset / Account Locked

Severity: High | Est. Resolution: 5–10 minutes | Affects: All staff | Likely Causes: Forgotten password, multiple failed attempts, stale cached credentials, or compromised account

#	Action	Details / Instructions
1	Verify Identity	Before any action, confirm identity via employee ID, security question, or registered mobile number using the company's MFA system.
2	Check Account Status in AD	Open Active Directory Users & Computers (ADUC). Search for the user. Check if 'Account is locked out' is shown. Note the lockout timestamp.

#	Action	Details / Instructions
3	Unlock the Account	In ADUC: right-click user > Properties > Account tab > tick 'Unlock account' > Apply. Or run in PowerShell: Unlock-ADAccount -Identity username
4	Reset the Password	Right-click user in ADUC > Reset Password. Set a temporary password (min. 10 chars: upper + lower + number + symbol). Tick 'User must change password at next logon'.
5	Clear Cached Credentials	Go to Credential Manager (Control Panel > Windows Credentials). Remove stale domain entries to prevent old passwords triggering repeat lockouts.
6	Communicate to User	Send the temporary password via SMS or phone call — never via unencrypted email. Instruct the user to log in and change it immediately.
7	Investigate Lockout Cause	Check Windows Security Event Log (Event ID 4625, 4740) on the domain controller. Common causes: mobile email app with old password, mapped drive, or scheduled task.
8	Document the Incident	Log in ManageEngine ServiceDesk: user, time, lockout source, and resolution steps. Flag if a security breach is suspected.

5.3 Scenario 3 — Slow Computer Performance

Severity: Low | Est. Resolution: 30–60 minutes | Affects: Any department | Likely Causes: High CPU/RAM usage, malware, too many startup programs, full disk, outdated drivers, overheating

#	Action	Details / Instructions
1	Check Resource Usage	Open Task Manager (Ctrl+Shift+Esc) > Performance tab. If CPU, RAM, or Disk is consistently above 85%, identify the top process in the Processes tab.
2	Close Unnecessary Programs	Ask the user which apps they need. Close everything else, including hidden system tray apps like Teams, OneDrive, or antivirus scanners running updates.
3	Restart the Computer	A restart clears RAM, applies pending Windows updates, and fixes many temporary slowdowns. Ensure the user saves all work first, then re-test.
4	Disable Startup Programs	Task Manager > Startup tab. Disable non-essential apps (Spotify, Skype, Steam). Restart and re-test.
5	Run Disk Cleanup	Open Disk Cleanup, select C: drive, tick all boxes including System files. Delete temp files, old updates, and Recycle Bin. Ensure at least 15% free disk space.
6	Scan for Malware	Run a full scan using Microsoft Defender or CrowdStrike. Quarantine any threats and report infections to the security team immediately.
7	Check Windows Updates	Settings > Windows Update. Install all pending updates and restart. Outdated systems run slower due to unpatched bugs.
8	Check Hardware Health	Use HWMonitor to check CPU temp (under 85 deg C under load). Run Windows Memory Diagnostic for RAM. Use CrystalDiskInfo for HDD health — flag bad sectors.
9	Recommend Hardware Upgrade	If the PC is over 5 years old, under 8 GB RAM, or running an HDD, document the specs and submit a hardware upgrade request to the IT Manager.

5.4 Escalation Policy

Tier	Who Handles	Scope	Escalate After
Tier 1	IT Helpdesk	Password resets, printer issues, basic software problems	30 minutes

Tier	Who Handles	Scope	Escalate After
Tier 2	IT Engineers	Network issues, hardware failures, complex software, AD/Server	2 hours
Tier 3	Senior IT / Vendor	Server outages, security incidents, ERP/core system failures	4 hours